

AdMu — Google Ads API Tool Design Document

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Company: Seitaiin Michibiki (整体院導) | Tool URL: <https://admu-backend-jxi0.onrender.com>

MCC Account ID: 167-626-3182 | API Contact Email: seitaimichibiki@gmail.com

1. Executive Summary

AdMu (Advertisement Mu) is a **fully developed and deployed** B2B SaaS platform that provides independent chiropractic and physical therapy clinics in Japan with an easy-to-use advertising management dashboard. The tool is live and serving clients at <https://admu-backend-jxi0.onrender.com>.

Current Status: AdMu is a **completed, production-deployed tool** with a fully functional backend, frontend dashboard, multi-tenant architecture, authentication system, and automated monitoring. The tool is currently operating in mock-data mode while awaiting API token approval — the only remaining step before connecting to live Google Ads data.

What AdMu does:

- **Reads** campaign performance data (impressions, clicks, cost, conversions) from each clinic's Google Ads account via the official SDK
- **Displays** the data in a simplified dashboard designed for non-technical clinic owners
- **Uploads** offline conversions (patient inquiry form submissions captured by gclid) back to Google Ads to measure real-world patient acquisition
- **Generates** AI-powered ad copy suggestions and negative keyword recommendations

What AdMu does NOT do:

- Does NOT create or modify campaigns, ad groups, or ads programmatically
- Does NOT perform automated bidding or budget changes via the API
- Does NOT access any personally identifiable information (PII) of ad viewers

2. Tool Category & Access Level

Item	Details
Tool Type	Internal tool used by the company to manage advertising for client clinics
Category	Reporting & Dashboarding + Offline Conversion Upload
Requested Access Level	Basic Access

Why Basic Access is needed	AdMu is a multi-tenant SaaS serving multiple clinic clients. Each clinic connects their own Google Ads account via OAuth 2.0. Test Account access only permits access to a single test account and cannot fetch real campaign data from paying clients' accounts.
Estimated daily API operations	~200–500 (well within Basic Access limits of 15,000/day)

3. Detailed API Usage

3.1 Google Ads API Services Used

API Service	Method	Read/Write	Purpose	Frequency
GoogleAdsService	SearchStream (GAQL)	READ	Fetch campaign-level metrics: impressions, clicks, cost_micros, conversions, CTR, CPC, CVR	On dashboard load + scheduled (every 5 min)
GoogleAdsService	SearchStream (GAQL)	READ	Fetch keyword-level and search term performance for negative keyword analysis	Weekly automated scan + on-demand
GoogleAdsService	SearchStream (GAQL)	READ	Fetch date-segmented performance data for trend charts (7/14/30/365 days)	On dashboard load
CustomerService	ListAccessibleCustomers	READ	Enumerate linked customer accounts under MCC for account selection	During initial OAuth connection
ConversionUploadService	UploadClickConversions	WRITE	Upload offline conversions (patient form submissions) linked by gclid to measure real patient acquisition from ads	Event-driven (on form submission)

3.2 Example GAQL Queries

```
-- Campaign performance (dashboard)
SELECT campaign.id, campaign.name, campaign.status,
       campaign_budget.amount_micros,
       metrics.impressions, metrics.clicks, metrics.ctr,
```

```

        metrics.average_cpc, metrics.cost_micros,
        metrics.conversions, metrics.conversions_from_interactions_rate
FROM campaign
WHERE segments.date DURING LAST_7_DAYS

-- Daily performance trend
SELECT segments.date, metrics.impressions, metrics.clicks, metrics.ctr,
        metrics.average_cpc, metrics.cost_micros, metrics.conversions,
        metrics.conversions_from_interactions_rate
FROM campaign
WHERE segments.date DURING LAST_30_DAYS
ORDER BY segments.date

-- Search term analysis (negative keyword detection)
SELECT search_term_view.search_term,
        campaign.id, campaign.name,
        metrics.clicks, metrics.cost_micros, metrics.conversions
FROM search_term_view
WHERE segments.date DURING LAST_30_DAYS
    AND metrics.impressions > 0
ORDER BY metrics.cost_micros DESC
LIMIT 500

```

3.3 Offline Conversion Upload Flow

1. Patient visits the clinic's website via a Google Ads click → `gclid` is captured in the URL
2. Patient submits an inquiry form → `gclid` + form data are stored in AdMu's database
3. AdMu backend calls `ConversionUploadService.UploadClickConversions` to send the conversion back to Google Ads
4. The conversion appears in the clinic's Google Ads reports as an offline conversion, enabling accurate ROI measurement

3.4 Data Retrieved and Storage

- **Data accessed:** Campaign/keyword aggregate metrics only (impressions, clicks, cost, conversions). No PII of ad viewers is accessed.
- **Storage:** Metrics are stored in PostgreSQL for up to 90 days for trend analysis, then automatically purged.
- **Data isolation:** Each clinic's data is isolated by `clinic_id` (tenant key). Cross-tenant access is prevented at the application layer.

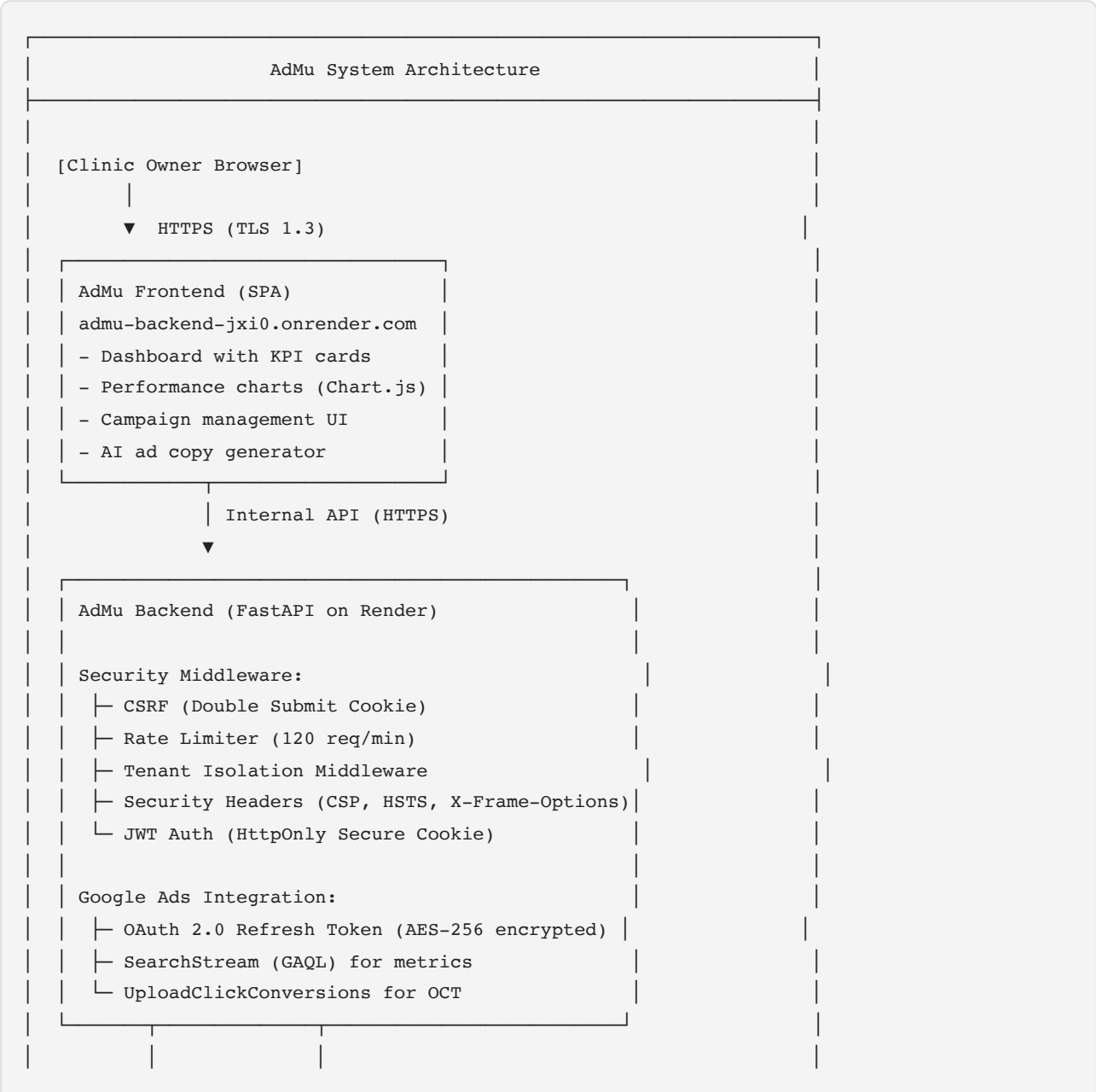
4. System Architecture

4.1 Technical Stack

Component	Technology	Details
Backend API	Python 3.11, FastAPI	RESTful API server with async support

Google Ads Client	google-ads-python SDK v25.1.0	Official Google Ads API client library
Database	PostgreSQL (Render.com managed)	Production database with automatic backups
Frontend	HTML5, Vanilla JavaScript, CSS	Responsive SPA dashboard
Hosting	Render.com (HTTPS enforced)	Auto-scaling with health checks
Authentication	OAuth 2.0 + JWT sessions	HttpOnly Secure cookies, CSRF protection
Monitoring	Sentry	Error tracking for both backend and frontend
Payments	Stripe	Subscription billing for SaaS clients
Notifications	LINE Messaging API + SMTP	Automated alerts and reports
AI Features	Google Gemini API	Ad copy generation and report analysis

4.2 Architecture Diagram





4.3 Automated Scheduler (Background Jobs)

Job	Frequency	Description
Campaign Health Check	Every 5 minutes	Monitors CTR/CVR drops, budget depletion. Sends LINE/email alerts.
Bid Rule Evaluation	Every 1 hour	Evaluates user-defined bid adjustment rules against current metrics.
Daily Report	9:00 AM JST daily	Sends yesterday's performance summary to clinic owner via LINE.
Weekly Report	Monday 8:00 AM JST	Sends 7-day performance summary via LINE + email.
Negative KW Scan	Wednesday 3:00 AM JST	Auto-detects wasted search terms (high cost, zero conversions) and adds to negative list.
Data Cleanup	Sunday 3:00 AM JST	Purges logs/alerts older than 365 days.

5. User Workflow

- Signup:** Clinic owner registers on AdMu. Account is created in "pending" state until admin approval.
- Admin Approval:** AdMu administrator reviews and activates the account. Welcome email is sent automatically.
- OAuth Connection:** Clinic owner connects their Google Ads account via standard Google OAuth 2.0 consent screen. AdMu receives a refresh token, which is AES-256 encrypted before storage.
- Dashboard:** On each visit, AdMu calls the Google Ads API (SearchStream/GAQL) using the stored refresh token to fetch campaign metrics. Data is displayed as interactive charts and KPI cards.
- Monitoring:** Background scheduler continuously monitors campaign health and sends alerts via LINE and email when anomalies are detected.
- Offline Conversion:** When a patient submits an inquiry form on the clinic's website, the captured gclid is sent to Google Ads as an offline conversion.

6. Security & Credential Management

6.1 Authentication & Authorization

Security Feature	Implementation
Session Management	JWT stored in <code>HttpOnly</code> , <code>Secure</code> , <code>SameSite=Lax</code> cookies. No localStorage token storage.
CSRF Protection	Double Submit Cookie pattern. All mutating API calls require X-CSRF-Token header matching the <code>csrf_token</code> cookie.
Rate Limiting	120 requests/minute per IP (general API), 3 requests/5 minutes for contact forms (bot protection).
Multi-tenant Isolation	Middleware validates <code>clinic_id</code> in every API request against the authenticated user's assigned clinic. Cross-tenant access returns HTTP 403.
Role-based Access	Two roles: <code>admin</code> (full system access) and <code>user</code> (clinic-scoped access only).

6.2 Credential Security

Credential	Protection Method
OAuth Refresh Tokens	AES-256-CBC encrypted at rest in PostgreSQL. Encryption key stored as server-side environment variable (<code>ENCRYPTION_KEY</code>).
Developer Token	Stored as server-side environment variable (<code>MASTER_ADS_DEVELOPER_TOKEN</code>). Never exposed to frontend.
Client Secret	AES-256 encrypted in database. Decrypted only in server memory during API calls.
User Passwords	bcrypt hashed (no plaintext storage).

6.3 Security Headers

- `Content-Security-Policy` — Restricts script/style sources to trusted CDNs only
- `Strict-Transport-Security` — `max-age=31536000; includeSubDomains`
- `X-Frame-Options: DENY` — Clickjacking prevention
- `X-Content-Type-Options: nosniff`
- `X-XSS-Protection: 1; mode=block`
- `Referrer-Policy: strict-origin-when-cross-origin`

7. Demo Access for Reviewers

Live Demo URL: <https://admu-backend-jxi0.onrender.com>

Demo Login Credentials:

Email: `demo@admu.jp`

Password: `demo2026admu`

Note: The demo account operates in mock-data mode (clearly labeled with a "● Mock Mode" badge in the UI). All dashboard features, charts, campaign views, AI ad copy generation, and negative keyword management are fully functional and can be explored. Once the developer token is approved, the system will seamlessly switch to live Google Ads data by configuring each tenant's OAuth credentials.

8. Compliance & Data Handling

- **No PII:** Only aggregate advertising metrics are accessed from the Google Ads API. No end-user personal data is retrieved.
- **Data Minimization:** Only the specific fields required for dashboard display are requested via GAQL SELECT statements.
- **Retention Policy:** Performance metrics are retained for 90 days for trend analysis, then automatically purged. Audit logs and alerts are purged after 365 days.
- **Tenant Isolation:** Each clinic's data is stored with a `clinic_id` foreign key. Cross-tenant access is prevented at both the middleware and database query levels.
- **Google API Terms Compliance:** AdMu does not resell raw Google Ads data. Data is used solely to generate the in-app dashboard for each account owner.
- **OAuth Consent:** Each clinic owner explicitly grants access via Google's standard OAuth consent screen before any data is read.
- **Token Revocation:** Clinic owners can disconnect their Google Ads account at any time from the settings page, which deletes the stored refresh token.

9. Company Information

Field	Details
Company Name	Seitain Michibiki (整体院導)
Business Type	B2B SaaS / Software Development for Healthcare Industry
Country	Japan
Company Website	https://michibiki-seitai.com
Tool URL (Live)	https://admu-backend-jxi0.onrender.com
API Contact Email	seitaimichibiki@gmail.com
Google Ads MCC ID	167-626-3182

10. Summary — Why Basic Access is Required

AdMu is a fully completed and deployed multi-tenant SaaS tool that needs to read live campaign performance data from each client clinic's Google Ads account. The tool:

1. Is **live and deployed** at <https://admu-backend-jxi0.onrender.com>
2. Has a **complete, production-ready backend** with PostgreSQL, security middleware, and automated monitoring
3. Uses the **official google-ads-python SDK** with proper GAQL queries already implemented
4. Has **AES-256 encryption** for all stored OAuth tokens
5. Implements **enterprise-grade security** (CSRF, rate limiting, CSP, HSTS, tenant isolation)
6. Serves **multiple client clinics** (multi-tenant), making Test Account access insufficient

The **only thing preventing live operation** is the developer token approval. All code, infrastructure, and security measures are in place.